Unit-IV

IoT Communication Protocols

4.1 Messaging Protocols

- Messaging protocols in IoT (Internet of Things) are crucial for enabling communication between IoT devices, gateways, and cloud platforms.
- These protocols define how data is transmitted, formatted, and interpreted across the IoT ecosystem.
- Several messaging protocols are used in IoT, each suited for different requirements and constraints.
- Here are some common messaging protocols in IoT:
    - MQTT (Message Queuing Telemetry Transport)
    - CoAP (Constrained Application Protocol)
    - XMPP (Extensible Messaging Presence Protocol)

4.1.1 MQTT

- MQTT is a standards-based messaging protocol, or set of rules, used for machine-to-machine communication.
- Smart sensors, wearables, and other Internet of Things (IoT) devices typically have to transmit and receive data over a resource-constrained network with limited bandwidth.
- These IoT devices use MQTT for data transmission, as it is easy to implement and can communicate IoT data efficiently.
- MQTT supports messaging between devices to the cloud and the cloud to the device.

MQTT Client and Broker

- Clients publish messages on different topics to brokers.
- The broker is the central server that receives these messages and filters them based on their topics.
- It then sends these messages to respective clients that have subscribed to those different topics.
- The heart of any publish/subscribe protocol is the MQTT broker.

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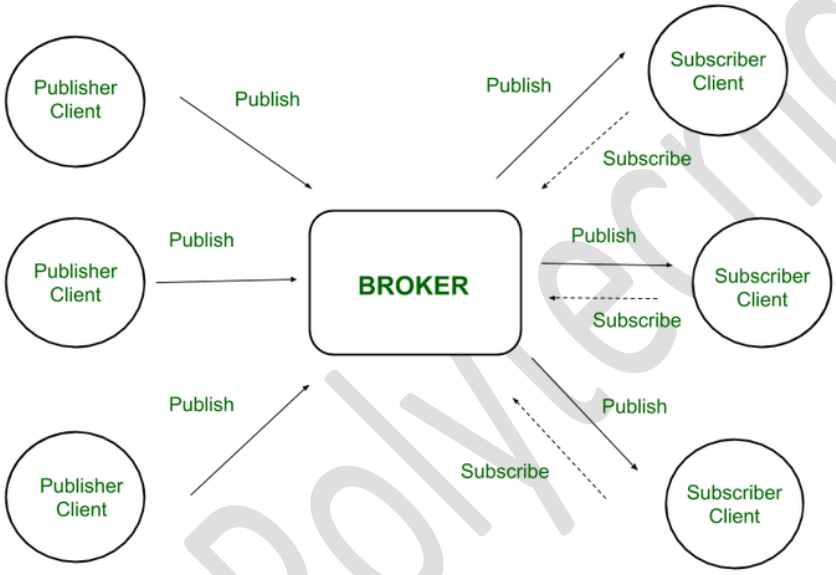
- A broker can handle up to thousands of concurrently connected MQTT customers, depending on how it is implemented.
- All communications must be received by the broker, who will then sort them, ascertain who subscribed to each one, and deliver the messages to the clients who have subscribed.
- All persistent customers' sessions, including missed messages and subscriptions, are likewise kept by the Broker.



- Hence client that has subscribed to a specific topic receives all messages published on that topic.
- Here the broker is central hub that receives messages, filters them, and distributes them to appropriate clients, such that both message publishers, as well as subscribers, are clients.

Working of MQTT

- MQTT's publish/subscribe (pub/sub) communication style, which aims to maximise available bandwidth, is an alternative to conventional client-server architecture that communicates directly with an endpoint.
- In contrast, the client who transmits the message (the publisher) and the client or clients who receive it (the subscribers) are not connected in the pub/sub paradigm.
- Third parties—the brokers—manage the relationships between the publishers and subscribers because they don't communicate with one another directly.

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- Publishers and subscribers, which denote whether a client is publishing messages or has subscribed to receive messages, are examples of MQTT clients.
- The same MQTT client can be used to accomplish these two features.
- A publish occurs when a client or device want to submit data to a server or broker.
- The term "subscribe" refers to the reversal of the procedure.
- Several clients can connect to a broker under the pub/sub paradigm and subscribe to subjects that interest them.

Advantages of MQTT

Lightweight and efficient

- MQTT implementation on the IoT device requires minimal resources, so it can even be used on small microcontrollers.
- For example, a minimal MQTT control message can be as little as two data bytes.
- MQTT message headers are also small so that you can optimize network bandwidth.

Scalable

- MQTT implementation requires a minimal amount of code that consumes very little power in operations.
- The protocol also has built-in features to support communication with a large number of IoT devices. Hence, you can implement the MQTT protocol to connect with millions of these devices.

Reliable

- Many IoT devices connect over unreliable cellular networks with low bandwidth and high latency.
- MQTT has built-in features that reduce the time the IoT device takes to reconnect with the cloud.

Secure

- MQTT makes it easy for developers to encrypt messages and authenticate devices and users using modern authentication protocols, such as OAuth, TLS1.3, Customer Managed Certificates, and more.

Well-supported

- Several languages like Python have extensive support for MQTT protocol implementation.
- Hence, developers can quickly implement it with minimal coding in any type of application.

4.1.2 CoAP

- **CoAP (Constrained Application Protocol)** is a session layer protocol that provides the RESTful (HTTP) interface between HTTP client and server.
- It is designed by IETF Constrained RESTful Environment (CoRE) working group.
- It is designed to use devices on the same constrained network between devices and general nodes on the Internet.
- CoAP enables low-power sensors to use RESTful services while meeting their low power constraints.
- This protocol is specially built for IoT systems primarily based on HTTP protocols.
- This network is used within the limited network or in a constrained environment.
- The whole architecture of CoAP consists of CoAP client, CoAP server, REST CoAP proxy, and REST internet.

Features of CoAP

- This protocol works extremely well in lower bandwidth. Successful communication over the IoT demands billions of nodes at a time. With its lower power consumption and low network overhead, it can run smoothly through such high congestion.
- CoAP has the ability to continue working even when TCP-based protocols like MQTT fail to communicate. In fact, it allows devices working with a poor signal to send their messages reliably. It can also enable an orbiting satellite to maintain successful distant communication.

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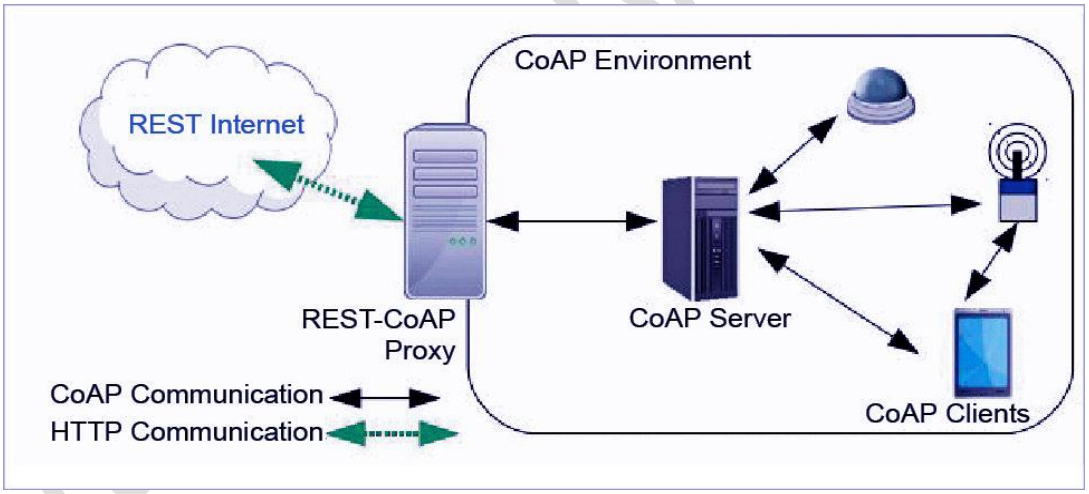
- CoAP works seamlessly with data formats like JSON, XML and more. Moreover, using cross-protocol proxies one can easily pair CoAP with various kinds of applications
- It has the ability to work with several microcontrollers while using the least possible resources. In fact, RAM space as low as 10 KB and code space as low as 100 KB is more than enough to run CoAP seamlessly.
- As the CoAP demands minimum resources, it keeps wastage under control. For instance, the header and encoding used for messaging are efficient and compact. Thus, fragments on the link layer are successfully avoided.

CoAP Architecture

- The data is sent from CoAP clients (such as smartphones, RFID sensors, etc.) to the CoAP server and the same message is routed to REST CoAP proxy.
- The REST CoAP proxy interacts outside the CoAP environment and uploads the data over REST internet.



- CoAP functions as a sort of HTTP for restricted devices, enabling equipment such as sensors or actuators to communicate on the IoT.
- These sensors and actuators are controlled and contribute by passing along their data as part of a system.
- The protocol is designed for reliability in low bandwidth and high congestion through its low power consumption and low network overhead.
- In a network with a lot of congestion or limited connectivity, CoAP can continue to work where TCP-based protocols such as MQTT fail to exchange information and communicate effectively.

Difference between COAP and MQTT protocols

MQTT	CoAP
This model has publishers and subscribers as main participants	Uses requests and responses
Central broker handles message dispatching, following the optimal publisher to client path.	Message dispatching happens on a unicasting basis (one-to-one). The process is same as HTTP.
Event-oriented operations	Viable for state transfer
Establishing a continual and long-lasting TCP connection with the broker is essential for the client.	Involved parties use UDP packets (async) for message passing and communication.
No message labeling but have to use diverse messages for different purposes.	It defines messages properly and makes its discovery easy.

4.1.3 XMPP

- Extensible Messaging and Presence Protocol (XMPP) is a communications protocol for message-oriented middleware based on XML (Extensible Markup Language).
- The protocol was originally named Jabber, and was developed by the Jabber open-source community in 1999. Later on, Jabber was formalized as the XMPP
- This protocol is mostly used by instant messaging applications like WhatsApp.

XMPP Architecture

- XMPP is a short form for Extensible Messaging Presence Protocol. It's protocol for streaming XML elements over a network in order to exchange messages and presence information in close to real time.

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- This protocol is mostly used by instant messaging applications like WhatsApp.
- Each character of word **XMPP**:
    - **X :** It means eXtensible. XMPP is a open source project which can be changed or extended according to the need.
    - **M :** XMPP is designed for sending messages in real time. It has very efficient push mechanism compared to other protocols.
    - **P :** It determines whether you are online/offline/busy. It indicates the state.
    - **P :** XMPP is a protocol, that is, a set of standards that allow systems to communicate with each other.
- These are the basic requirements of any Instant Messenger which are fulfilled by XMPP:
    1. Send and receive messages with other users.
    2. Check and share presence status
    3. Manage subscriptions to and from other users.
    4. Manage contact list
    5. Block communications(receive message, sharing presence status, etc) to specific users.

XMPP Advantages:

- It is free and decentralized which means anyone can set up an XMPP server.
- It is based on open standards.
- It supports multiple implementations of clients and servers.
- It is flexible, XML-based and can be extended. So, suitable for both instant messaging features and custom cloud services.
- Security is supported via SASL and TLS.
- It is efficient, can support million of concurrent users on a single service such as GTalk.

4.2 Transport Protocols

- Transport protocols define the rules and mechanisms for transmitting data between devices or systems over a network.
- They handle tasks such as segmentation, reassembly, error detection, and flow control, ensuring reliable and efficient data transfer.

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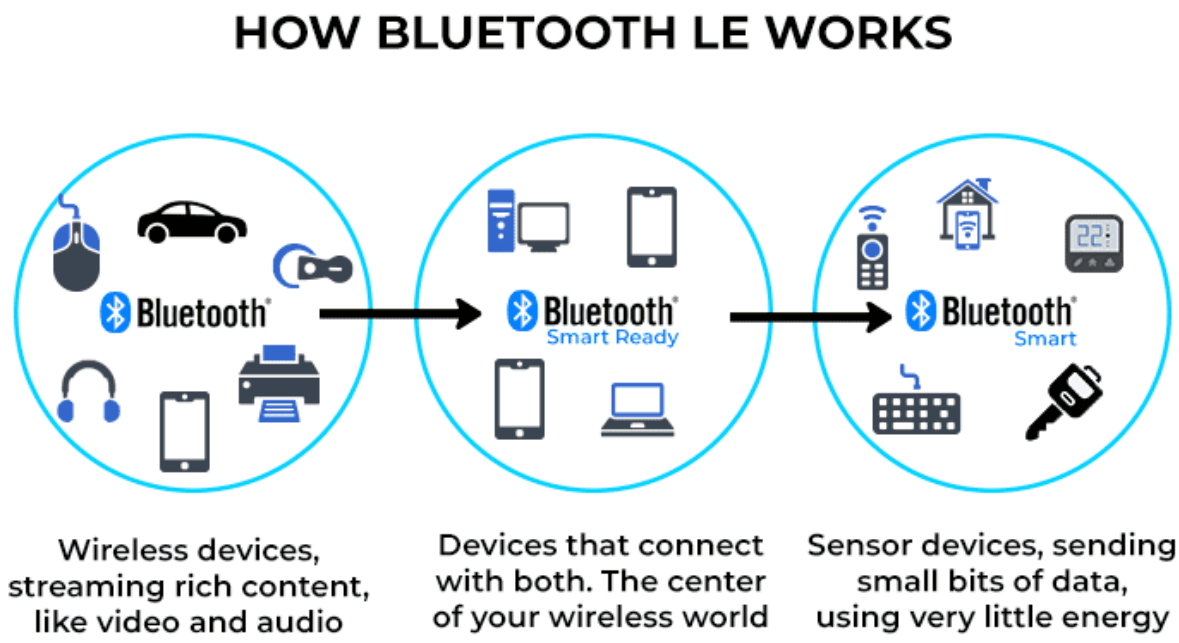
4.2.1 Introduction of BLE

- **Bluetooth Low Energy (BLE)** is similar to Bluetooth classic.
- It is a short-range wireless LAN communication technology standardized by IEEE 802.15.
- It also operates in 2.4 GHz ISM unlicensed bandwidth.
- The major distinguishing feature of BLE, commercially made accessible first in 2011, is its low power consumption while maintaining the same communication range as in Bluetooth classic.
- It typically finds its use in healthcare, fitness, security and home automation appliances that save power.



Advantages

Low power consumption

- Even when compared to other low-power technologies.
- BLE achieves optimized and low power consumption by keeping the radio off as much as possible and sending small amounts of data at low transfer speeds.

Low cost of entry into development

- BLE modules and chipsets are low cost when compared to other similar technologies.

- This is a result of the increased adoption rate and competition in the marketplace.

Open access to documentation

- With most other wireless protocols and technologies, you would have to become a member of the official group or consortium for that standard to access the specification.

Prevalence in smartphones

- This is probably the biggest advantage BLE has over its competitors, such as ZigBee, Z-Wave, and Thread.
- The vast majority of people in the world own a smartphone, and almost all of those smartphones have BLE hardware inside.
- This gives developers a much larger potential user base for their applications.

Disadvantages

1. **Limited Range**: BLE has a relatively short range compared to other wireless technologies like Wi-Fi or classic Bluetooth. Typically, the range is around 10 meters, which can be a limitation for certain applications requiring longer distances.

2. **Limited Bandwidth**: BLE is designed for low-power applications, which means it has a lower data transfer rate compared to classic Bluetooth. This limitation can be problematic for applications requiring high-speed data transmission.

3. **Compatibility**: BLE compatibility varies across different devices and versions of the Bluetooth standard. Ensuring compatibility between devices from different manufacturers or different BLE versions can sometimes be challenging.

4. **Security Concerns**: While BLE includes security features such as encryption and authentication, vulnerabilities have been identified in the past. Weaknesses in the implementation of security protocols or outdated firmware can expose BLE devices to security risks such as eavesdropping or unauthorized access.

5. **Complexity of Development**: Developing BLE applications can be more complex compared to other wireless technologies due to the need to manage

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power consumption efficiently, handle connection states, and deal with potential interference issues.

Applications

- **Wearable Technology:** BLE has become a main protocol in the development of wearable devices, powering smartwatches, fitness trackers, and health monitoring devices. Its low power consumption is applied in ensuring these devices can operate for extended periods without frequent recharging.

- **Healthcare Devices:** BLE finds maximum use in healthcare applications, connecting medical devices to smartphones for real-time monitoring. This includes devices like blood glucose monitors, heart rate monitors, and temperature sensors.

- **Smart Home Devices:** BLE is a key enabler of smart home ecosystems. From smart bulbs and thermostats to door locks and security systems, BLE provides seamless communication between devices, enhancing the overall smart home experience.

- **Asset Tracking:** BLE is used for asset tracking applications, and businesses to monitor the location of assets within a limited space. This is valuable in industries such as logistics, where real-time tracking can streamline operations.

4.2.2 Introduction to Li-Fi

- LIFI, also known as LIGHT FIDELITY, is a both-sided wireless communication system that releases information via LED (Light Emitting Device) or Infrared Light. LIFI technology was the first time it came into use in the year 2011.
- LIFI is an extension of WIFI Communication which requires a light source for data sending and communication on the internet.
- The most significant feature of LIFI is that it's most faster than WIFI communication.
- LiFi increases the speed and bandwidth of wifi, 3G, and 4G. LiFi transmission speeds can go over 100 Gbps, 14 times faster than WIFI Communication, also known as the world's fastest wifi.

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- The interference or the obstacles are less during the communication process, so in seawater, it is preferable to use LIFI as WIFI communication can have numerous interference while communicating.

Working of Li-Fi

1. The working or the implementation of LIFI is not very complex like WIFI communication. It is straightforward to send information through the LED light bulb, which emits pulsed light to receivers. After that, the receiver collects the sent data or information and understands the transmitted data.
2. The LED bulb can be toggled on and off instantly for more reliable communication and for saving power. So only the things we require for this communication are some LEDs Bulbs and a controller that codes into those LEDs Bulbs.
3. In some cases, we need to alter or modify the rate of transmission in which the lED bulb flash depending upon the information we want to encode.
4. As we stated above, the Light source is the main component in this communication, so that light source will serve as the central hub for data transmission from one point to another destination point.

Advantages of Li-Fi

- **High Speeds:** Li-Fi can achieve data transfer speeds significantly faster than traditional Wi-Fi. Theoretically, Li-Fi can achieve speeds of several gigabits per second, making it suitable for high-bandwidth applications.
- **Security:** Since light cannot penetrate through walls like radio waves, Li-Fi offers inherent security advantages by confining the signal within a specific area. This makes it more secure against interception and hacking.
- **Availability:** Light sources are ubiquitous in indoor environments, making Li-Fi readily available for deployment in various settings such as offices, hospitals, and homes.
- **Interference-Free:** Unlike Wi-Fi, which operates in crowded radio frequency bands, Li-Fi operates in the unregulated light spectrum, minimizing interference from other wireless devices.

Disadvantages of Li-Fi

- LIFI has a minimal area of scope or range also connectivity.
- There is an Unavailability of Compatible Technologies.

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- LIFI can give rise to Light Interference and Light Pollution.
- LiFI has Possible cost Implications.
- Cannot penetrate through walls.
- Interference from external sources i.e. sun, normal bulbs, candles, etc

Applications:

- **Indoor Wireless Communication:** Li-Fi can be used for indoor wireless communication in environments such as offices, hospitals, museums, and retail stores.
- **High-Speed Data Transfer:** Li-Fi can be deployed in scenarios requiring high-speed data transfer, such as multimedia streaming, video conferencing, and augmented reality applications.
- **Secure Environments:** Li-Fi's inherent security features make it suitable for use in secure environments such as military facilities, government offices, and financial institutions.

Difference between Li-Fi and Wi-Fi

Key	LiFi	WiFi
Technology Used	Present IrDA compliant devices	WLAN 802.11a/b/g/n/ac/ad standard-compliant devices
Privacy	In LiFi, light does not pass through the walls and hence will provide a much more secure data transfer	In WiFi, RF signal passes through the walls and hence there is a need to employ techniques to achieve secure data transfer.
Speed	10 Gbps	150 Mbps
Interference	Less Interference and Obstacles while Communication	High Interference and Obstacles while Communication
Data density	Works in the high dense environment	Works in the less dense environment due to interference-related issues

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Cost	Less Cost (No extra Requirements)	High Cost (Due to Complex Structure)
Application	Underwater communication, EMI-sensitive environments, Dense urban environments.	wi-fi hotspot.
Data transfer speed,	About 1 Gbps WLAN-11n offers 150Mbps	About 1-2 Gbps can be achieved using WiGig/Giga-IR
Area	LiFi covers 10 meters of area.	WiFi covered 30 meters of area.
Standard	IEEE 802.15.7	IEEE 802.11

4.3 Basics of Sensor Network Topologies

- Sensor network topologies refer to the ways in which sensors are interconnected within a network.
- Here's a list of common sensor network topologies:
    - Point to Point Topology
    - Star Topology
    - Mesh Topology
    - Ring Topology

4.3.1 Point to Point Topology

- In a point-to-point topology, also known as a peer-to-peer topology, each sensor node is directly connected to another sensor node, forming a network where every node can communicate only with its directly connected neighbors.
- There is no central hub or intermediary device involved in the communication between nodes.

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Advantages:

1. **Simplicity**: P2P networks are straightforward to set up and maintain because there are no intermediary devices or complex routing protocols involved. Each node only needs to establish connections with its direct neighbors.
2. **Low Latency**: Communication between nodes in a P2P network typically has low latency since there are no intermediate devices involved in routing messages. This makes P2P networks suitable for applications requiring real-time or low-latency communication.
3. **Scalability**: P2P networks can be scalable to a certain extent. Adding new nodes to the network typically involves establishing direct connections between the new node and its neighbors. However, managing connections may become complex as the number of nodes increases.
4. **Reliability**: In terms of reliability, P2P networks can be robust if the individual connections between nodes are stable. Failure of one node generally doesn't affect the communication between other nodes in the network.

Disadvantages:

1. **Limited Range**: P2P networks are typically used for short-range communication between neighboring nodes. As a result, they may not be suitable for applications requiring long-distance communication without additional infrastructure such as repeaters or signal boosters.
2. **Scalability Challenges**: While P2P networks can be scalable to some extent, managing connections and addressing potential bottlenecks may become challenging as the network grows larger. This can lead to increased complexity and potential performance issues.
3. **Single Points of Failure**: Although failure of one node generally doesn't affect communication between other nodes in the network, a failure in a critical node or connection could disrupt communication between specific pairs of nodes, leading to potential single points of failure.
4. **Limited Redundancy**: P2P networks may lack built-in redundancy compared to other network topologies like mesh networks. If a direct connection between two nodes fails, there may not be alternative routes for communication, leading to potential communication disruptions.

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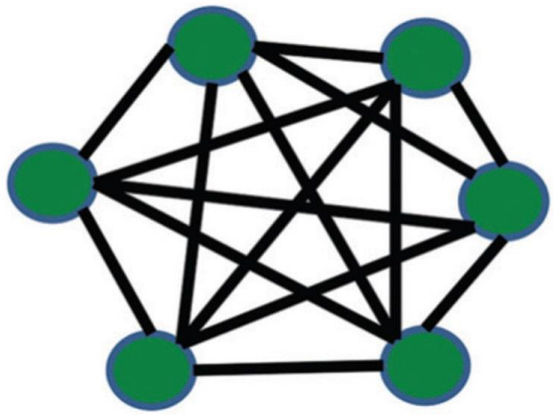
4.3.2 Mesh topology

- Mesh topologies consist of interconnected sensors where each sensor node can communicate with multiple neighboring nodes.
- Data can be routed through multiple paths, providing redundancy and fault tolerance.
- Mesh topologies offer robustness and scalability, making them suitable for large-scale IoT deployments where reliability is critical.



Advantages:

1. **Redundancy and Fault Tolerance**: Mesh networks offer multiple paths for data to travel from the source to the destination. This redundancy enhances fault tolerance; if one node or connection fails, data can still find alternative routes to reach its destination, improving the overall reliability of the network.
2. **Scalability**: Mesh networks are highly scalable. New nodes can be added to the network easily, and each node can communicate directly with multiple neighboring nodes. This allows for the expansion of the network without significant changes to the overall topology.
3. **Self-Healing**: Mesh networks are self-healing in nature. If a node or connection fails, the network can dynamically reroute data through alternative paths, bypassing the failed component. This self-healing capability helps to maintain continuous operation even in the presence of failures.
4. **Coverage and Range**: Mesh networks can provide extensive coverage and range, especially when nodes act as relays to forward data between distant nodes. This makes mesh topologies suitable for IoT applications that require wide area coverage or communication over long distances.

5. **Load Balancing**: Mesh networks can distribute traffic across multiple paths, enabling load balancing and optimizing network performance. This can help prevent congestion in certain areas of the network and ensure efficient utilization of resources.

Disadvantages:

1. **Complexity**: Mesh networks can be more complex to design, deploy, and manage compared to simpler topologies like star or point-to-point. The need for routing protocols and dynamic routing algorithms adds complexity to the network infrastructure.
2. **Overhead**: Mesh networks may incur overhead due to the need for routing protocols, routing tables, and additional control messages for managing network topology and routing paths. This overhead can consume network bandwidth and processing resources.
3. **Latency**: While mesh networks offer redundancy and fault tolerance, the presence of multiple hops between nodes can introduce latency in data transmission. Longer paths or congested routes may result in increased latency, which can impact real-time applications that require low latency communication.
4. **Power Consumption**: In battery-powered IoT devices, acting as relays to forward data between distant nodes in a mesh network can consume more power compared to simpler topologies where nodes communicate directly with a central hub. This increased power consumption can affect the battery life of individual nodes.

4.3.3 Ring topology

- Ring topologies involve sensors connected in a circular manner, forming a closed loop.
- Each sensor is connected to two neighboring sensors, except for the end sensors, which are connected to one sensor each.
- Ring topologies offer uniform data transmission delay but are less common in IoT deployments due to their susceptibility to single point failures.

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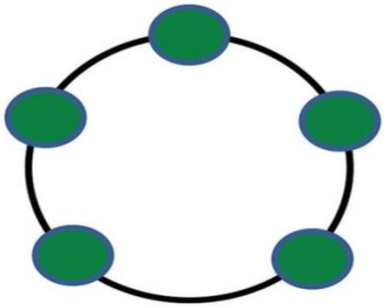

Advantages:

1. **Simplicity**: Ring topologies are relatively simple to set up and understand. Each node is connected to exactly two other nodes, forming a closed loop. This simplicity can make ring networks easy to deploy and manage, especially in smaller-scale deployments.
2. **Equal Access**: In a ring topology, each node has equal access to the network, as there are no central hubs or switches that may introduce bottlenecks or uneven traffic distribution. This can result in more predictable performance and fairer resource allocation.
3. **Fault Isolation**: Ring topologies offer built-in fault isolation. If one node or connection fails, data can still circulate around the ring in the opposite direction, bypassing the failed component. This can help contain network failures and minimize their impact on overall network operation.
4. **Deterministic Performance**: Ring networks can offer deterministic performance, as data travels in a predictable, sequential manner around the ring. This can be advantageous for applications requiring strict timing or synchronization, such as industrial automation or control systems.

Disadvantages:

1. **Single Point of Failure**: Despite offering fault isolation, ring topologies are vulnerable to a single point of failure. If the connection between two nodes is severed or if a node fails, the entire network can become fragmented, disrupting communication for all nodes in the ring.
2. **Limited Scalability**: Ring topologies may have limited scalability compared to other topologies like mesh or star. Adding new nodes to the network may require reconfiguration of existing connections, and the overall performance of the network may degrade as the number of nodes increases.
3. **Limited Bandwidth**: In a ring topology, the available bandwidth is shared among all nodes in the ring. As a result, the total bandwidth available to

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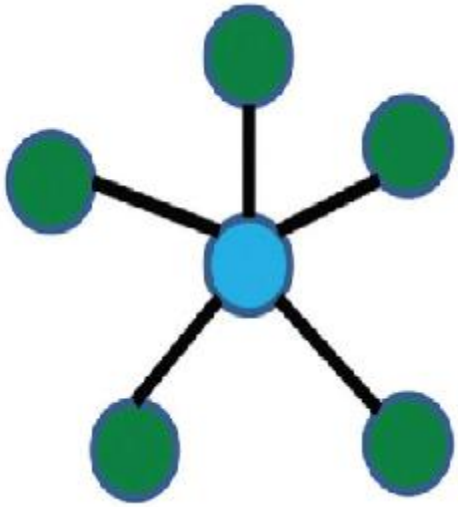
each node may be limited, especially as the number of nodes increases or if there is heavy traffic on the network.

4. **Latency**: Data transmission in a ring topology involves passing through each node in sequence, which can introduce latency, especially if the ring is large or if there are congestion points in the network. This latency may be acceptable for some applications but may be prohibitive for real-time or latency-sensitive applications.

4.3.4 Star Topology

- In sensor networking, a star topology involves connecting all sensor nodes directly to a central hub or gateway device.
- Each sensor node communicates exclusively with this central hub, which collects, processes, and manages the sensor data.
- This setup offers centralized data collection, simplifies network management, and enables efficient communication between sensor nodes and the central hub.
- However, it introduces a single point of failure—the central hub—which can disrupt communication if it fails.
- Despite this risk, a star topology is commonly used for its simplicity, scalability, and centralized control in sensor networking applications.



Advantages:

1. **Simplicity**: Star topologies are simple to set up and manage. All sensors are connected directly to a central hub or switch, which facilitates easy installation and troubleshooting. This simplicity makes star topologies suitable for small to medium-sized deployments.

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2. **Scalability**: Star topologies are inherently scalable. New sensors can be added to the network by connecting them directly to the central hub or switch, without affecting the existing network infrastructure. This scalability makes star topologies flexible and adaptable to changing requirements.
3. **Centralized Management**: In a star topology, the central hub or switch acts as a central point of control and management for the network. This centralized management simplifies network administration tasks such as configuration, monitoring, and security enforcement.
4. **Fault Isolation**: Star topologies offer built-in fault isolation. If one sensor fails or experiences issues, it does not affect the operation of other sensors in the network. This fault isolation makes it easier to identify and troubleshoot problems in the network.
5. **High Performance**: Star topologies can offer high performance, especially in networks with high-speed central hubs or switches. Each sensor has dedicated bandwidth to communicate with the central hub, minimizing congestion and ensuring efficient data transmission.

Disadvantages:

1. **Single Point of Failure**: Despite offering fault isolation, star topologies are vulnerable to a single point of failure—the central hub or switch. If the central device fails, communication between all sensors in the network can be disrupted, leading to a complete network outage.
2. **Limited Range**: Star topologies are typically limited by the range of the central hub or switch. Sensors must be located within the coverage area of the central device, which may restrict the geographical spread of the network, especially in large-scale deployments.
3. **Dependency on Central Device**: All communication in a star topology relies on the central hub or switch. If the central device becomes overloaded or experiences issues, it can degrade network performance and affect the operation of all sensors connected to it.
4. **Bandwidth Bottleneck**: In star topologies, all sensors share the bandwidth of the central hub or switch. If multiple sensors need to transmit data simultaneously, congestion can occur at the central device, leading to performance degradation and increased latency.

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